

DIRISA Student Datathon Challenge 2026 | Team Eduvos

Hung Council Watch

Exploratory data analysis, machine learning and the live dashboard

Which South African municipalities are most likely to experience a hung council after the 2026 Local Government Elections, and how can two decades of political fragmentation data be used to forecast where coalition government will be needed?

Live dashboard: eduvos-hung-councils.netlify.app

Five elections, one linked dataset

We combined every local government election since 2000 into one table that follows each municipality through 21 years of boundary changes.

5

elections: 2000, 2006, 2011,
2016 and 2021

213

municipalities today, every
one covered

263

councils linked across
boundary and code changes

What we measured, from the PR (party) ballot

- **Leading party's share:** above 50% means a majority.
- **Winning margin:** the gap between first and second place.
- **Turnout:** the share of registered voters who voted.
- **Fragmentation:** the Effective Number of Parties, which counts parties by their size.

Effective Number of Parties

Twenty parties on the ballot means little if one of them wins 90%. This measure weights each party by its vote share.

One party wins 90%	1.2
Two parties split 50 / 50	2.0
Three equal parties	3.0

Cleaning that changed the results

We restored 22 Western Cape municipalities lost to Excel's row limit, and proved that the 2016 turnout column counted only the leading party's votes. Every fix is in a cleaning log.

One-party dominance has collapsed

70% → 16%

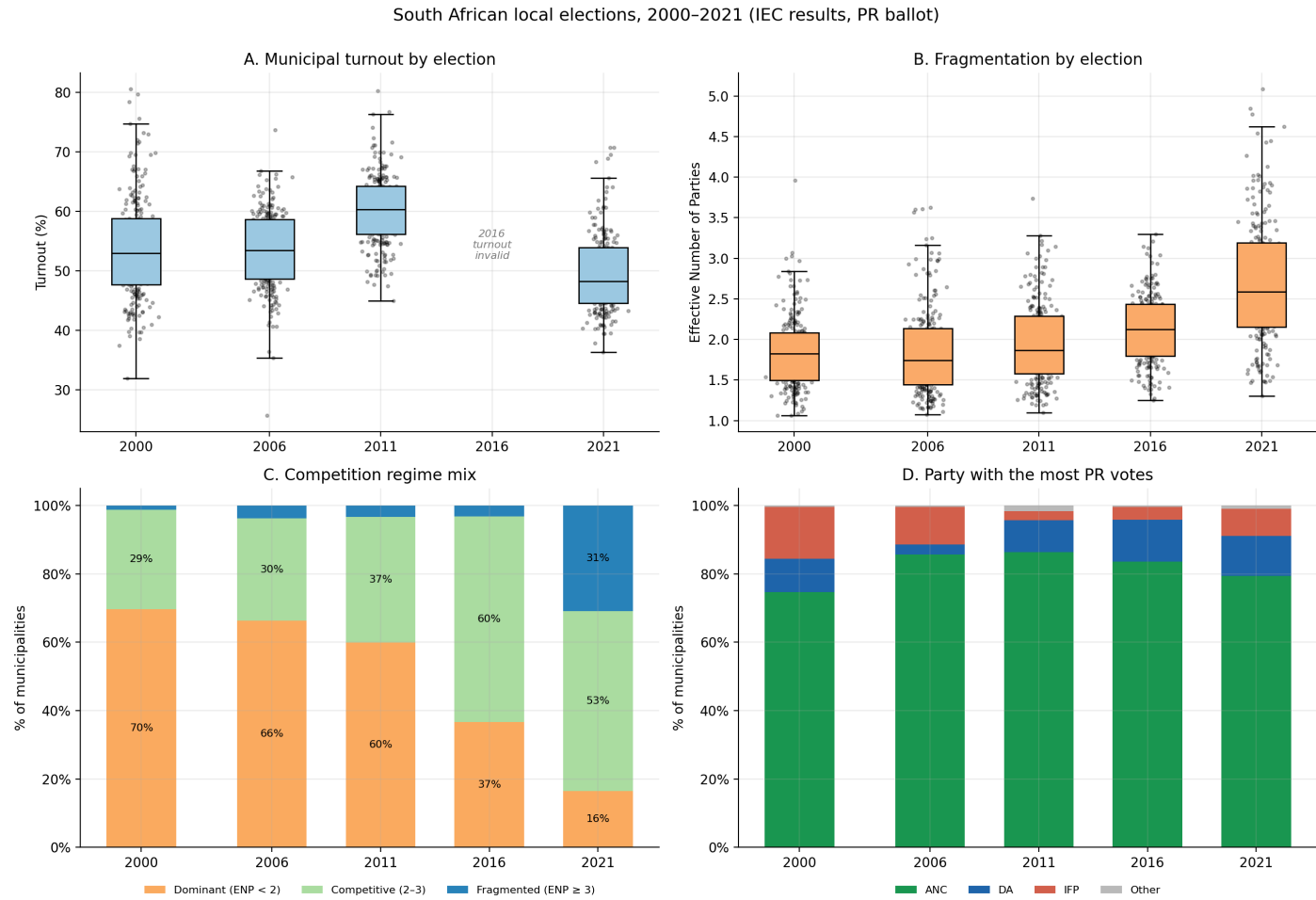
of municipalities dominated by one party
(fewer than 2 effective parties), 2000 to
2021

1% → 31%

highly fragmented (3 or more effective
parties)

27 → 72

hung councils after 2016 and after 2021:
more than double in one election



District Management Areas excluded. 2016 turnout excluded (invalid in source file).

Our first hypothesis: does turnout follow an inverted U?

We expected turnout to be low in one-party towns, highest in battlegrounds and low again in fragmented councils. We tested it with the Lind–Mehlum U-test in each election.



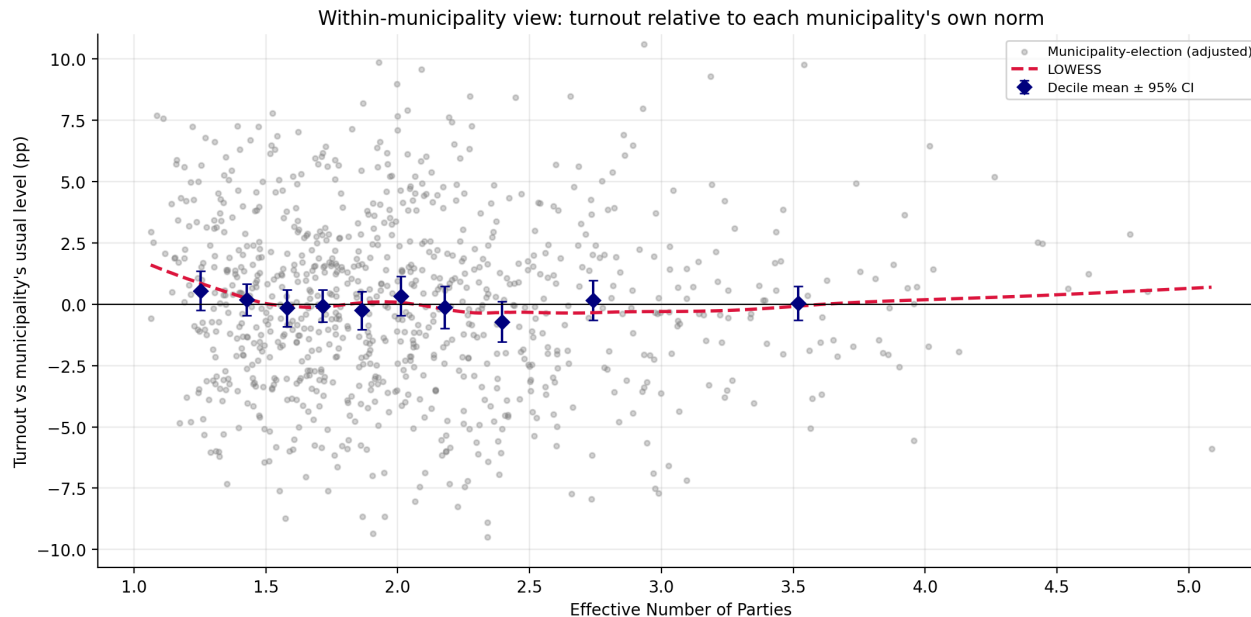
Election	2000	2006	2011	2021
Inverted-U test p-value	0.06	0.96	0.42	0.42
Supported? ($p < 0.05$)	No	No	No	No

Not supported in any election

An early version found the U-shape in 2021. It disappeared once the missing Western Cape municipalities were restored.

Following the same municipality over time

The strongest test: when a municipality became more fragmented, did its own turnout change? We removed each council's usual level and each election's national swing (fixed effects).



< 1%

of a municipality's turnout changes explained by fragmentation
(inverted-U $p = 0.98$)

-10.8 pts

median turnout drop 2011 to 2021, almost everywhere, whether
fragmentation rose or not

Turnout moves with the **national mood** and **regional differences**, not with competition.

What the EDA told us, and where it led

We tested

An inverted U between turnout and competition, across five elections and within municipalities.



It was rejected

Turnout is driven by national swings and regions. Incomplete data had briefly created a false pattern.



The data showed

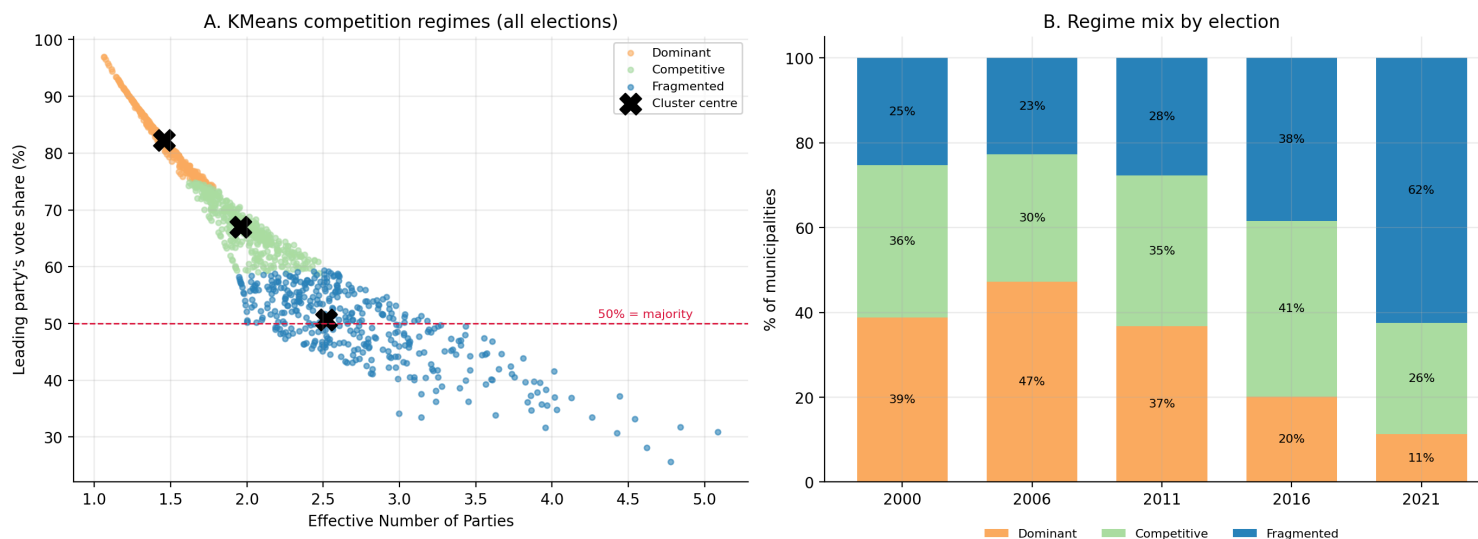
One-party dominance collapsing, and hung councils more than doubling in 2021.

So we asked the question that matters for 4 November 2026: **which councils are most likely to be hung, and where will coalition government be needed?**

Testing our first idea honestly, and letting the evidence change our question, is the strongest result of the EDA.

Two models, two different jobs

Our first prototype trained XGBoost to predict KMeans labels built from the same features. It scored 99% while learning nothing new (target leakage). We separated the jobs.



KMeans describes

Groups councils into Dominant, Competitive and Fragmented from vote shares only. Stable across 20 random starts. The Fragmented group sits on the 51% line and grew from 25% to 62% of councils.

A supervised model forecasts

Predicts whether each council will be hung at the **next** election from what we know now, so the answer is never part of the input.

Tested like an exam: 2021 was hidden until the end



Model	Chosen on: validation, in time order	2021 test ROC-AUC
Baseline (no learning)	0.44	0.88
Simple logistic regression	0.44	0.88
Full logistic regression	0.42	0.72
XGBoost (tuned)	0.34	0.79

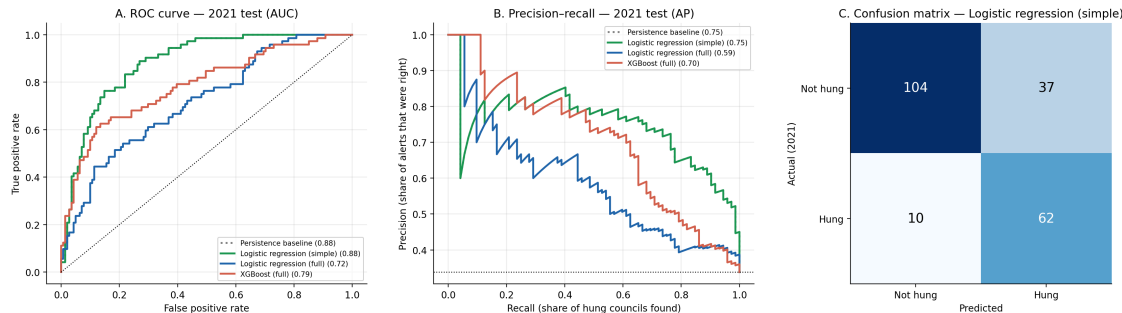
Validation score: PR-AUC on 2011 and 2016, learning only from earlier elections. Cross-validation grouped by municipality was used to tune XGBoost.

62 of 72

councils that became hung in 2021 flagged in advance (ROC-AUC 0.88)

Why the simple model won

XGBoost looked best in ordinary cross-validation, but its patterns did not carry over between elections. How close the leader is to 50% held up every time.



The 2026 forecast

77

hung councils expected across 213 municipalities
(72 after 2021)

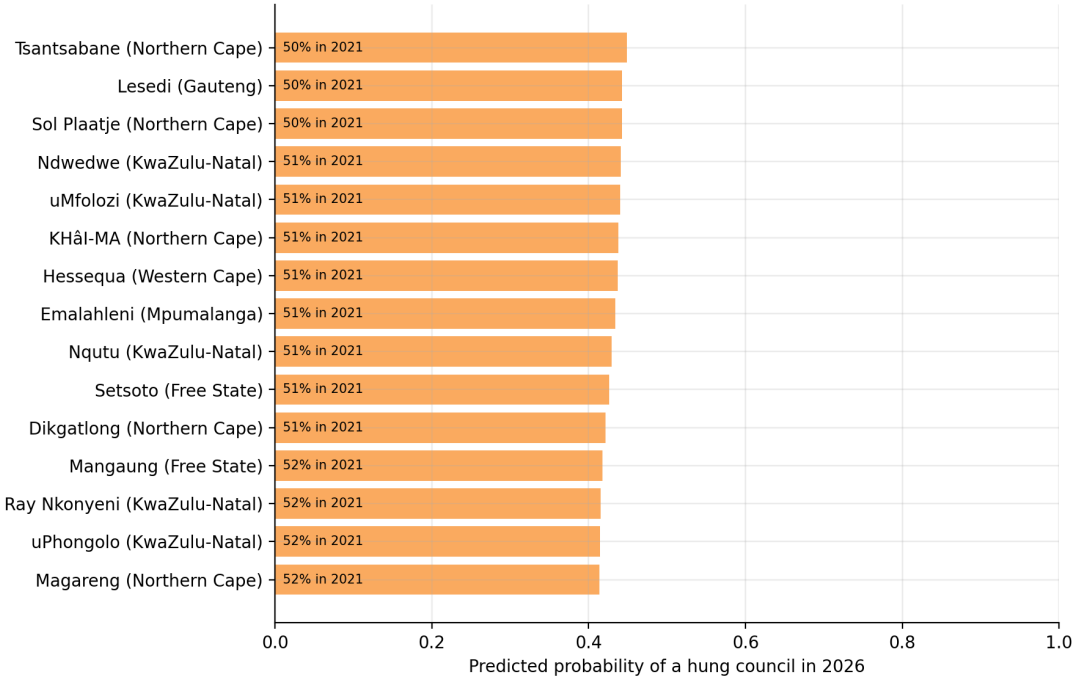
61

at high risk, 72 battlegrounds, 80 safe

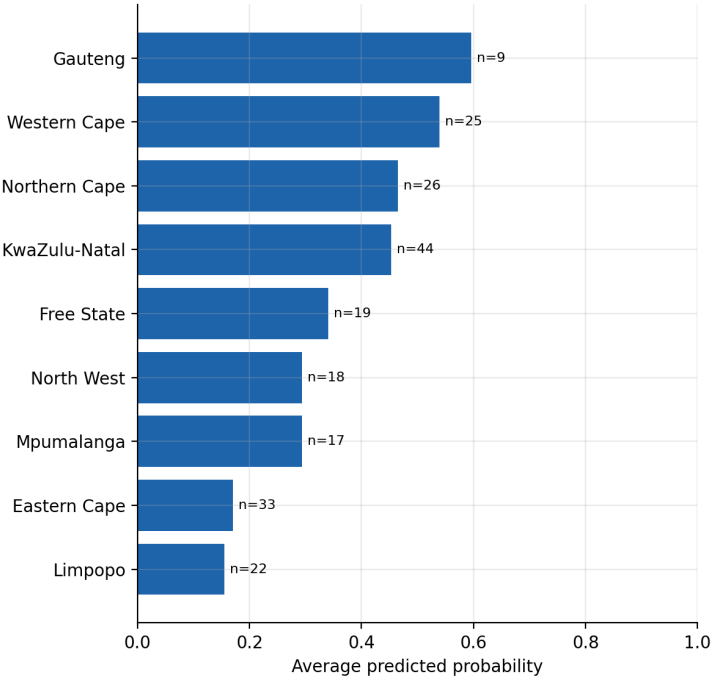
~100

if every leading party loses another 6 points, as it did from 2016 to 2021

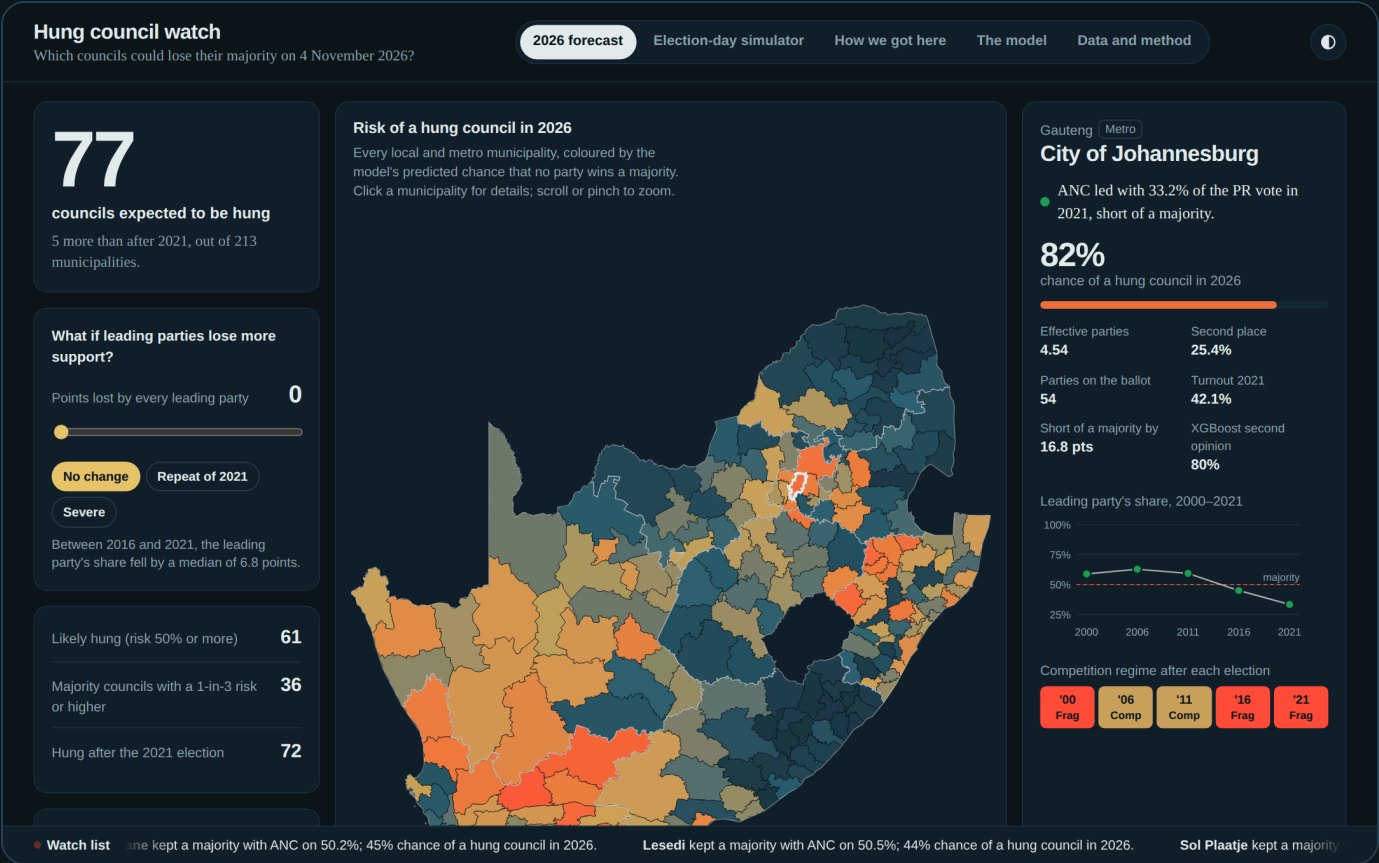
A. Watch list: majority councils most at risk of becoming hung



B. Average risk by province



Hung Council Watch, live for anyone to use



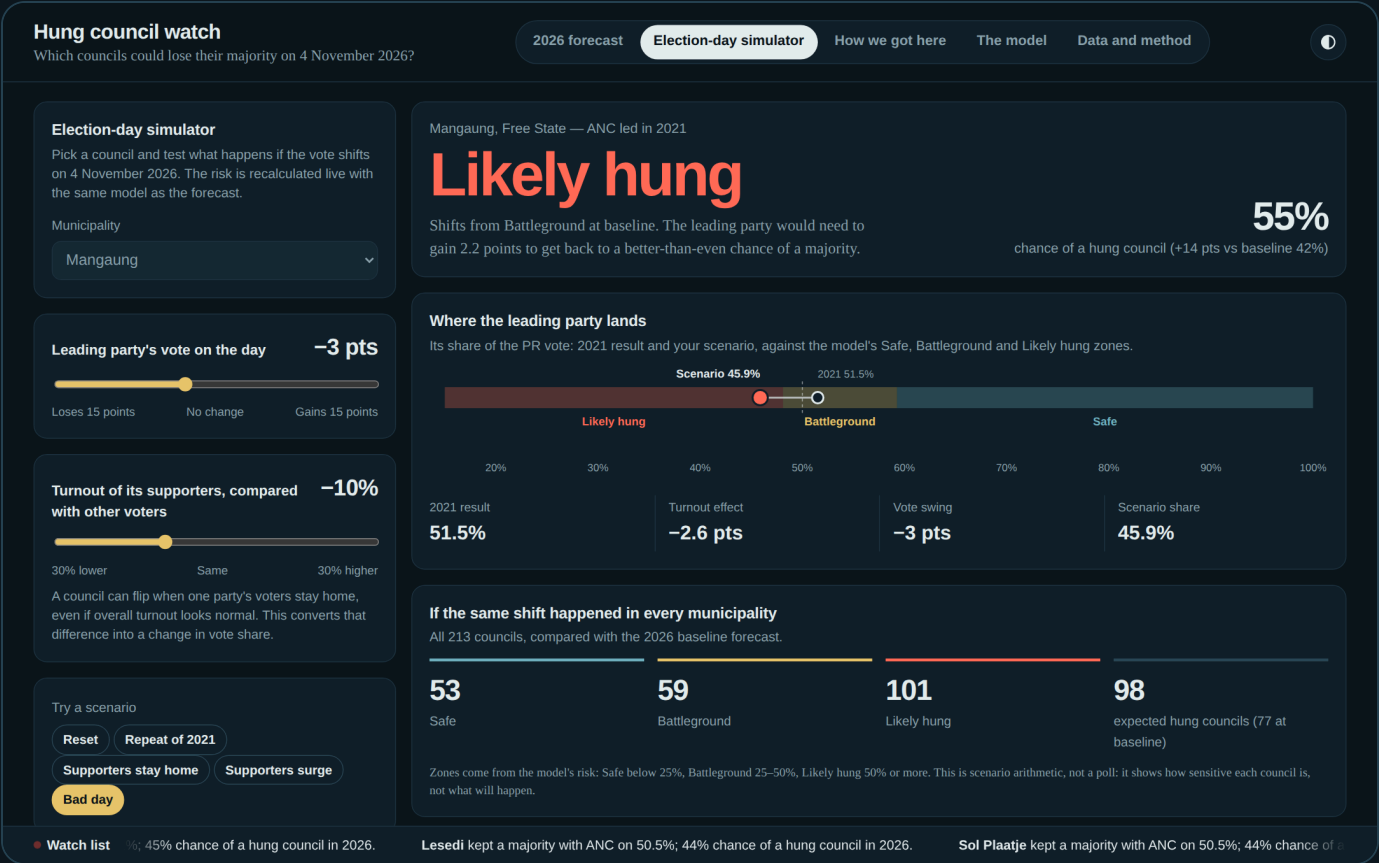
Five tabs

- 2026 forecast: risk map of all 213 councils, swing slider, watch list
- Election-day simulator
- How we got here: 2000–2021 time-lapse
- The model and Data and method



Scan or visit
eduvos-hung-councils.netlify.app

Election-day simulator: test any council



Built to be reliable, honest and easy to use

The model runs in the browser

Every slider recalculates each council's risk with the trained model's own coefficients. No server, no delay, nothing to fall asleep.

One self-contained file

Data, official boundaries and code in a single page, hosted free on Netlify and working even offline.

Honest by design

No plain turnout slider, because turnout does not drive the result in our analysis. The simulator models who stays home instead.

Traceable numbers

Every figure matches the notebooks, so the dashboard, code, slides and video tell the same story.



Try it yourself: eduvos-hung-councils.netlify.app